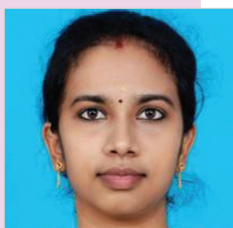


Simple Tricks To Pack MORE 5G ANTENNAS

Can compact isolation structures enable denser 5G antennas? PCB-friendly isolation techniques make it possible to place more antennas into tight layouts without sacrificing isolation, gain, or efficiency.



DR K. MITHRA is an independent RF and wireless researcher based in Scottsdale, Arizona, with a PhD in Information & Communication Engineering. She has worked extensively on compact antenna arrays, 5G performance optimisation, and secure wireless transmission.

Designers of 5G radios face a constant squeeze: product teams demand more antennas in less space to support higher capacity, beamforming, and MIMO, while RF physics penalises elements placed too close together. As spacing shrinks, mutual coupling quietly degrades matching, distorts radiation patterns, and reduces array efficiency.

Across antenna research and consulting projects for 3.5GHz 5G systems, the outcome often depends on how effectively coupling is controlled within just a few millimetres of PCB space. Small layout decisions in this narrow gap frequently determine whether an array performs reliably or falls short.

Compact isolation structures, such as parallel-coupled line resonators and carefully shaped ground features, provide a practical way to increase antenna density in 5G and future systems. These solutions fill the limited gaps already available, use standard PCB processes, and deliver measurable isolation improvements in real hardware.

Why spacing alone is not enough

The simplest way to reduce coupling is to increase the distance between elements. For many 5G arrays at 3.5GHz, designers aim for element spacings around half a wavelength. That approach works until mechanical constraints, size limitations, and multi-band packaging push antennas closer together.

Once you move below these comfortable spacings, following happens:

- Surface waves on the substrate become a significant coupling path.
- The input impedance of each element changes due to its neighbours.
- Radiation patterns and front-to-back ratio degrade, directly impacting coverage and MIMO performance.

In consulting work, I have seen arrays that simulated well as isolated elements but failed in the final product because real-world spacing and housing forced much tighter layouts. At that point, we can no longer ‘buy’ isolation with space; we must engineer the coupling path itself.

There are many published solutions:

defected ground structures (DGS), electromagnetic bandgap (EBG) patterns, complementary split-ring resonators, metamaterial-inspired structures, and even vertical isolation walls. These approaches work, but they often require extra board area, non-planar structures, or complex geometries that may not be suitable for cost-sensitive, high-volume hardware.

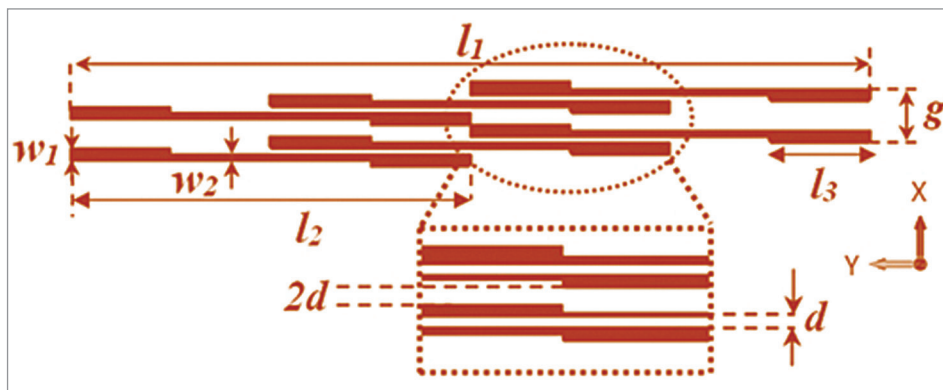


Fig. 1: Decoupling element